Michael Ritchot

Singapore · michael@ritchot.me · LinkedIn · GitHub · Website

SUMMARY

International educator with 11 years of classroom experience across four countries, spanning MYP Design, Mathematics, Computer Science, and Sciences. Built a parallel track in educational technology leadership: digital learning ecosystems, emergent technology integration, and staff professional development. Designs, builds, and evaluates learning systems end to end: designed and evaluated an asynchronous e-learning module teaching AI literacy through the mechanics of large language models, improving assessment scores 28 points pre-to-post (62% to 90%) through a Design-Based Research methodology, then extended the work into a four-module program delivered on a custom-engineered platform. Led school-wide technology implementations including an enterprise LMS rollout and an organization-wide knowledge management system scaled from a single-department pilot to mandated division-wide use. Experienced in constructing interdisciplinary, inquiry-driven learning experiences grounded in constructionist and behaviorist approaches, and in embedding generative AI into curriculum design, assessment, and academic integrity practice. Committed to purposeful technology integration, collaboration, and the continuous evolution of teaching and learning.

CORE COMPETENCIES

IB MYP Curriculum · Curriculum Design & Vertical Alignment · Assessment & Rubric Design · Data-Informed Instruction · Differentiation & EAL Scaffolding · Educational Technology Integration · Generative AI / LLMs · AI Literacy Instruction · LMS Implementation · Staff Professional Development · Instructional Design (ADDIE) · Inquiry & Project-Based Learning · Interdisciplinary Program Design · Academic Integrity · Learning Analytics · Design-Based Research

FEATURED PROJECT — AI LITERACY FOR THE MODERN WORKFORCE

- Designed a four-module, 37-section learning program with 12 practice activities, pre/post competency assessments, and adaptive sequencing, applying ADDIE methodology and Anthropic's 4D competency framework (Delegation, Description, Discernment, Diligence) as an organizing taxonomy.
- Engineered and deployed the delivery platform end to end (React, TypeScript, Tailwind CSS) on Cloudflare Pages, building interactive demonstrations of LLM mechanics — client-side BPE tokenization and real softmax-with-temperature probability visualizations — that make abstract model behavior directly observable to learners.
- Built the learning analytics layer: xAPI-aligned event tracking, active reading time measurement, and an administrator dashboard surfacing live learner data for instructional decision-making.
- Established a Kirkpatrick Levels 1–4 evaluation framework: a reaction-survey item bank, a ten-item parallel-form pre/post assessment, twelve behavioral transfer indicators on 30/60/90-day intervals, and a Phillips-methodology ROI model.
- Grounded the program in a structured needs analysis using action mapping and learner persona development, then delivered the full development cycle solo in approximately 210 hours — a 71% reduction against the Chapman Alliance industry baseline of roughly 735 hours for comparable Level-3 e-learning.

View the full program ↗ View the source code ↗

EXPERIENCE

MYP Design Teacher

Stamford American International School Singapore · Singapore · July 2025 – Present

- Applied data-informed teaching using formative assessments, process journals, and a post-unit survey system to guide differentiation and unit revision. Students returned a 95% "Encourages Perseverance" score on the 2026 Voice of the Student Survey.
- Led an AI-assisted redesign of three Grade 8 (MYP Year 3) Design units — Open Source Robotics, Scaled Architecture, and 3D Modeling — rebuilding e-journals, instructional decks, and rubrics from a single design specification with Claude Cowork. EAL vocabulary scaffolding runs throughout, improving curricular progression and assessment coverage. Configured a Flint AI tutor as a Socratic guide for students.
- Designed a cross-departmental AI literacy lesson for the IB Sports and Exercise Science team, translating principles from graduate research on LLM mechanics into a four-activity instructional sequence targeting critical evaluation of AI-generated content in Internal Assessment workflows. Produced a comprehensive facilitator guide enabling independent delivery by subject-area teachers without AI expertise.
- Served on the secondary curriculum inquiry cycle reviewing the Design and Computer Science programs school-wide, covering written curriculum examination, transition and performance data analysis, and integrated recommendations against Cognita and IB expectations. Contributed to vertical alignment of the Design curriculum under the Enhanced MYP framework.
- Consulted on the design and structure of a new AI Literacy co-curricular activity, advising the Secondary Digital Learning Coordinator on curriculum sequencing and instructional materials grounded in AI fluency frameworks.
- Engineered a streamlined back-end reporting architecture for the Design department to solve reporting cohesion and inefficiency. The system became widely adopted by peers, significantly reducing administrative overhead during reporting cycles.

Mathematics and Computer Science Teacher | Emergent Technology Coach

TIS Macao · Macao SAR, China · August 2020 – July 2025

- Taught Grade 8 – 9 Mathematics and Grade 10 – 12 Computer Science, following Alberta Ministry of Education and IB standards, achieving standardized scores 11–17% above the assessment average.
- Overhauled the Middle School Mathematics program to follow both Alberta and IB standards, and developed 13 interdisciplinary MYP Mathematics investigations for Grades 8 and 9, including designing task-specific assessment rubrics based on MYP Mathematics Criteria, and training on the usage of generative AI tools to assist with rubric creation. One investigation had students generate design ideas with AI tools and 3D-print scale architectural models.
- Developed and revised internal assessments for Grades 7 through 9 (MYP 2 through 4), including blueprint maps and mark schemes. Ensured assessments aligned with MYP Mathematics command terms, IB MYP content benchmarks, and the Alberta program of studies for a thorough and standardized evaluation process. Assessment data analysis confirmed internal grading aligned within 5% of external benchmarks.
- Led professional development workshops on Google Workspace productivity strategies, Generative AI, and VR/AR, which received widely positive feedback and resulted in self-reported workload improvements and stress reduction among participants, and provided new avenues for authentic emergent technology use for learning.
- Consolidated and reorganized Secondary Mathematics resources into a shared system that grew into a Google Workspace redesign, creating comprehensive informational documentation. This initiative enhanced both synchronous and asynchronous collaboration among teachers, improved transparency between grade levels, and provided a structured repository for resource continuity. These enhancements resulted in better internal benchmark tracking and improved retention of institutional knowledge, ultimately implemented across all of Secondary.
- Expanded a Mathematics competition program that provided further avenues for Mathematical learning, resulting in a 35% increase in participation over two years, with students earning accolades, recognition, and prizes.

Sciences and Literary Studies Teacher | Technology Coordinator and Coach

Shanghai Nanyang Model Private High School (CINEC) · Shanghai, China · August 2018 – July 2020

- Piloted and designed courses in Grade 12 Environmental Sciences, Grade 11 Life Sciences, and Grade 10 – 11 Literary Studies, following British Columbia Ministry of Education and IB standards.
- Led the feasibility study and full-scale implementation of Schoology, a school-wide LMS that centralized academic reporting, course delivery and improved communication between teachers, students, parents, and other stakeholders.
- Led professional development sessions on educational technology tools to improve teaching and learning based on upcoming research and integrative reviews.

Mathematics, Computer Science, Sciences, and Physical Education Teacher

SIS Canada · Sokcho, South Korea · August 2016 – July 2018

- Built the Middle School Computer Science curriculum, defining the scope, sequence, and assessment for Grades 7 – 9.
- Taught and designed courses in Grade 7 – 9 Mathematics, Computer Science, Sciences, and Physical Education, following British Columbia Ministry of Education standards.
- Designed STEM-focused curriculum for Grades 7–9 while establishing student-led digital media initiatives, including a video editing tech club and a social media marketing committee.

Mathematics Teacher

Canadian International Academy Shanghai · Shanghai, China · August 2015 – July 2016

- Taught and designed courses in Grade 10 – 12 Mathematics, following Ontario Ministry of Education Curriculum standards.

Operations Admin

TForce Integrated Solutions · Edmonton, Alberta, Canada · September 2010 – February 2015

- Developed a new cost analysis system for the Edmonton branch, which gained nationwide implementation due to its effectiveness in cost savings, ease of transference between branches, and adaptability to individual branch needs.
- Liaised between various branches and shipping companies to resolve shipping and payment disputes.
- Performed various administrative duties such as maintaining and recording audited accounts, broker invoices, payroll, and created cost analysis reports.

Office Clerk

Department of National Defence · Edmonton, Alberta, Canada · June 2009 – August 2010

- Performed a variety of duties, including implementing the National Household Removal Services contract, mediating disputes between clients and agents, preparing briefings to educate clients on their contracted rights, and entering payroll transactions.

EDUCATION

M.Ed. in Education Technology & Instructional Design

Western Governors University · Salt Lake City, UT, United States · 2025

- Received the Excellence Award in Learning Experience Design, a distinction granted to the top 1–2% of graduate students for outstanding academic and project performance.
- Designed and evaluated an asynchronous e-learning module teaching AI literacy through the mechanics of large language models (LLMs), improving secondary student assessment scores 28 points pre-to-post (62% to 90%); derived actionable design principles using a Design-Based Research methodology to advance K–12 AI literacy instruction.

Bachelor of Education

University of Alberta · Edmonton, Alberta, Canada · 2015

CERTIFICATIONS

Professional Credentials

- Alberta Ministry of Education Professional Certificate — Alberta, Canada · 2023
- BC Ministry of Education Professional Certificate of Qualification — British Columbia, Canada · 2016
- Human Subjects Research SBE Comprehensive — CITI Program · 2025

Teaching & Curriculum

- MYP Mathematics Category 1 — International Baccalaureate · 2022
- Google Certified Educator Level 1 and 2 — Google for Education · 2024
- Apple Certified Teacher — Apple · 2024

AI & Emerging Technology

- AI Fluency: Framework & Foundations — Anthropic · 2025
- Teaching the AI Fluency Framework — Anthropic · 2025
- AI Capabilities and Limitations — Anthropic · 2026
- AI Fluency for Educators — Anthropic · 2025
- AI Fluency for Nonprofits — Anthropic · 2025
- Claude 101 — Anthropic · 2026
- Introduction to Claude Cowork — Anthropic · 2026
- Foundations for Teachers — OpenAI · 2025
- AI Literacy Certified Level 1, Level 2, and Level 3 — Flint · 2026
- Google AI Essentials — Google · 2024
- Gemini Certified Educator — Google · 2025

Data Analytics & Project Management

- Google Data Analytics Professional Certificate — Coursera/Google · 2024
- Google Project Management Certificate — Coursera/Google · 2024